# Grade 10 – Term 1
# Comparing Decision Criteria: C8 Practice

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## Evaluated criteria

**C8** Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.

## 1 Introduction

All problems use **profit payoffs**: larger values are better, and negative values represent losses. Alternatives $A_i$ occupy rows; mutually exclusive states of nature $S_j$ occupy columns. The payoff $P_{ay}(A_i, S_j)$ is the profit if $A_i$ is chosen and $S_j$ occurs. A decision is made before the state is known.

### 1.1 Four ways to compare the same alternatives

**Maximax** selects the alternative with the highest row maximum. It asks, "Which alternative offers the greatest possible profit?" **Maximin** selects the alternative with the highest row minimum. It asks, "Which alternative has the best worst-case profit?" Neither needs probabilities, even when probabilities are available.

**Minimax Regret** first compares alternatives within each state:

$$b(S_j) = \max_i P_{ay}(A_i, S_j), \qquad r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

Regret is the profit forgone compared with the best alternative for the state that actually occurs; it is not necessarily a cash loss. Select the alternative with the *smallest maximum regret*. This criterion needs the full payoff table, but no probabilities.

**EMV/Bayes** selects the highest Expected Monetary Value:

$$EMV(A_i) = \sum_{j=1}^{n} P(S_j)P_{ay}(A_i, S_j), \qquad P(S_j) \geq 0, \quad \sum_{j=1}^{n} P(S_j) = 1.$$

It requires a probability for every state as well as the payoffs. In these problems the state probabilities are the same for every alternative. Maximizing expected money expresses a risk-neutral orientation: it does not directly penalize uncertainty or protect against a large loss. An EMV is a weighted average, not a guaranteed profit or necessarily an attainable single outcome.

| Criterion | Information and selection | Risk orientation | Main limitation |
|---|---|---|---|
| Maximax | Payoffs; highest row maximum | Optimistic; pursues upside | Ignores other outcomes and their likelihoods. |
| Maximin | Payoffs; highest row minimum | Conservative; protects the worst payoff | Ignores upside and how unlikely the worst state may be. |
| Minimax Regret | Full payoff table; smallest row maximum regret | Cautious about missed opportunities | Ignores likelihoods; depends on the alternatives included. |
| EMV/Bayes | Payoffs and valid probabilities; highest EMV | Risk-neutral about money | Sensitive to probability estimates; may recommend an unaffordable downside. |

Criteria can disagree because they compare different summaries of the *same* information. Agreement does not mean that their reasoning is identical. Report all tied alternatives; do not invent a tie-breaking rule. Without probability information, EMV cannot be calculated unless an additional probability assumption is explicitly justified. Equal probabilities are not implied by uncertainty.

**How to complete the numerical problems.** The summary tables are already completed; do not recalculate row maxima, row minima, regrets, or EMVs. Identify the recommended alternative(s) by comparing the appropriate summary column. In the answer table, record the winning summary value and name the quantity it represents (for example, "Row minimum: 35"). Then justify and compare the recommendations using the supplied summaries and the original payoffs.



## 2 Comparing Decision Criteria

**Problems:**


2.1. Worked comparison: delivery expansion

2.2. Festival catering

2.3. Retail plans: agreement and ties

2.4. Bakery production


### 2.1 Delivery Expansion

A delivery firm chooses its expansion plan. Quarterly profits are in thousand US dollars. Forecast probabilities are shown in the table.

| Alternative | Strong ($S_1$) | Stable ($S_2$) | Weak ($S_3$) |
|---|---|---|---|
| $P(S_j)$ | 0.20 | 0.60 | 0.20 |
| Large fleet ($A_1$) | 100 | 20 | -20 |
| Flexible fleet ($A_2$) | 75 | 50 | 10 |
| Current fleet ($A_3$) | 40 | 40 | 35 |
| Contract partners ($A_4$) | 60 | 65 | 20 |

Evaluating these payoffs and forecast probabilities under standard decision criteria yields the following summary statistics:

| Alternative | Row Maximum | Row Minimum | Maximum Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 100 | -20 | 55 | 28 |
| $A_2$ | 75 | 10 | 25 | 47 |
| $A_3$ | 40 | 35 | 60 | 39 |
| $A_4$ | 65 | 20 | 40 | 55 |

**Questions/Tasks**



**C8** Which alternatives does each criterion recommend? Complete the answer table.

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | _____ | _____ |
| Maximin | _____ | _____ |
| Minimax Regret | _____ | _____ |
| EMV/Bayes | _____ | _____ |

**Solution**

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_1$ | Row maximum: 100 |
| Maximin | $A_3$ | Row minimum: 35 |
| Minimax Regret | $A_2$ | Maximum regret: 25 |
| EMV/Bayes | $A_4$ | EMV: 55 |

Maximax selects $A_1$ (highest payoff: 100), despite a minimum of $-20$. Maximin selects $A_3$ (highest minimum: 35), sacrificing potential gains. Minimax Regret selects $A_2$ (lowest maximum regret: 25), but its minimum payoff is only 10. EMV/Bayes selects $A_4$ (highest EMV: 55), although its actual payoffs are 60, 65, and 20.

Maximax is optimistic; Maximin is conservative; Minimax Regret minimizes missed opportunities; EMV/Bayes maximizes expected profit using state probabilities. Only EMV uses probabilities, while regret depends on competing

alternatives. None guarantees a required profit. If avoiding losses is mandatory, exclude $A_1$ before applying any criterion.



## 2.2 Festival Catering

Use the completed summary below to identify all four recommendations in the blank answer table. A caterer chooses a large or small contract before knowing attendance. Profits are in thousand US dollars.

| Alternative | High ($S_1$) | Low ($S_2$) |
|---|---|---|
| $P(S_j)$ | 0.30 | 0.70 |
| Large contract ($A_1$) | 80 | 10 |
| Small contract ($A_2$) | 45 | 35 |

| Alternative | Row Maximum | Row Minimum | Maximum Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 80 | 10 | 25 | 31 |
| $A_2$ | 45 | 35 | 35 | 38 |

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | _____ | _____ |
| Maximin | _____ | _____ |
| Minimax Regret | _____ | _____ |
| EMV/Bayes | _____ | _____ |

### Questions/Tasks



**C8** Which alternatives does each criterion recommend? Complete the answer table.

### Solution

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_1$ | Row maximum: 80 |
| Maximin | $A_2$ | Row minimum: 35 |
| Minimax Regret | $A_1$ | Maximum regret: 25 |
| EMV/Bayes | $A_2$ | EMV: 38 |

Maximax selects the highest row maximum (80); Maximin selects the highest row minimum (35); Minimax Regret selects the smallest maximum regret (25); EMV/Bayes selects the highest EMV (38). There are no ties.



## 2.3 Retail Plans: Agreement and Ties

Use the completed summary below to identify all four recommendations in the blank answer table. A retailer compares three plans. Profits are in thousand CNY.

| Alternative | Strong ($S_1$) | Normal ($S_2$) | Weak ($S_3$) |
|---|---|---|---|
| $P(S_j)$ | 0.25 | 0.50 | 0.25 |
| Plan $A_1$ | 60 | 45 | 30 |
| Plan $A_2$ | 50 | 40 | 20 |
| Plan $A_3$ | 60 | 35 | 30 |

| Alternative | Row Maximum | Row Minimum | Maximum Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 60 | 30 | 0 | 45 |
| $A_2$ | 50 | 20 | 10 | 37.5 |
| $A_3$ | 60 | 30 | 10 | 40 |

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | _____ | _____ |
| Maximin | _____ | _____ |
| Minimax Regret | _____ | _____ |
| EMV/Bayes | _____ | _____ |

### Questions/Tasks



**C8** Which alternatives does each criterion recommend? Complete the answer table.

### Solution

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_1$ and $A_3$ | Row maximum: 60 |
| Maximin | $A_1$ and $A_3$ | Row minimum: 30 |
| Minimax Regret | $A_1$ | Maximum regret: 0 |
| EMV/Bayes | $A_1$ | EMV: 45 |

Maximax selects the highest row maximum (60), shared by $A_1$ and $A_3$. Maximin selects the highest row minimum (30), also shared by $A_1$ and $A_3$. Minimax Regret selects $A_1$ with the smallest maximum regret (0), and EMV/Bayes selects $A_1$ with the highest EMV (45). Both tied alternatives must be reported for Maximax and Maximin.



## 2.4 Bakery Production

Use the completed summary below to identify all four recommendations in the blank answer table. A bakery selects a daily production plan before demand is known. Profits are in hundred US dollars.

| Alternative | High ($S_1$) | Normal ($S_2$) | Low ($S_3$) |
|---|---|---|---|
| $P(S_j)$ | 0.30 | 0.50 | 0.20 |
| Large batch ($A_1$) | 90 | 30 | -10 |
| Flexible batch ($A_2$) | 65 | 50 | 20 |
| Small batch ($A_3$) | 40 | 40 | 35 |

| Alternative | Row Maximum | Row Minimum | Maximum Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 90 | -10 | 45 | 40 |
| $A_2$ | 65 | 20 | 25 | 48.5 |
| $A_3$ | 40 | 35 | 50 | 39 |

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | _____ | _____ |
| Maximin | _____ | _____ |
| Minimax Regret | _____ | _____ |
| EMV/Bayes | _____ | _____ |

### Questions/Tasks

**C8** Which alternatives does each criterion recommend? Complete the answer table.

### Solution

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_1$ | Row maximum: 90 |
| Maximin | $A_3$ | Row minimum: 35 |
| Minimax Regret | $A_2$ | Maximum regret: 25 |
| EMV/Bayes | $A_2$ | EMV: 48.5 |

Maximax selects the highest row maximum (90), despite a possible loss of 10. Maximin selects the highest row minimum (35). Minimax Regret and EMV/Bayes agree on $A_2$, using the smallest maximum regret (25) and highest EMV (48.5), respectively. There are no ties; only EMV uses the demand probabilities.

# 3 Contextual Practice

**Problems:**


3.1. Urban farm crop mix

3.2. Business reserve investment

3.3. Coastal ferry capacity

3.4. Printing workshop equipment


Use the supplied summaries for Task 1, reporting all four recommendations and deciding values, including ties. For Task 2, solve the threshold inequality and identify eligible alternatives. Update only affected summaries; compute regrets using eligible alternatives only. All monetary variables are in thousand US dollars. Probabilities remain unchanged.

## 3.1 Urban Farm Crop Mix

### Problem Description

An urban farm chooses a crop mix for the next season. Profits are in thousand US dollars; the estimates come from a market forecast.

| Alternative | Strong ($S_1$) | Stable ($S_2$) | Weak ($S_3$) |
|---|---|---|---|
| $P(S_j)$ | 0.50 | 0.30 | 0.20 |
| Specialty crops ($A_1$) | 80 | 40 | -10 |
| Balanced crops ($A_2$) | 60 | 55 | 20 |
| Staple crops ($A_3$) | 40 | 45 | 35 |

| Alternative | Row Maximum | Row Minimum | Maximum Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 80 | -10 | 45 | 50 |
| $A_2$ | 60 | 20 | 20 | 50.5 |
| $A_3$ | 45 | 35 | 40 | 40.5 |

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | _____ | _____ |
| Maximin | _____ | _____ |
| Minimax Regret | _____ | _____ |
| EMV/Bayes | _____ | _____ |

### Questions/Tasks



**C8** Which alternatives does each criterion recommend? Complete the answer table.

**C8** The farm requires a minimum profit of 20 in every market state. A government subsidy adds $s$ to $A_2$'s weak-market payoff, where $s$ is an integer between 0 and 20. Find the smallest $s$ that makes Maximin prefer $A_2$ over $A_3$. Justify numerically, identify eligible alternatives, and report all four restricted recommendations and deciding values at that subsidy.

**Solution**

1. The winning values in each summary column give:

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_1$ | Row maximum: 80 |
| Maximin | $A_3$ | Row minimum: 35 |
| Minimax Regret | $A_2$ | Maximum regret: 20 |
| EMV/Bayes | $A_2$ | EMV: 50.5 |

   There are no ties. Maximax favors $A_1$'s upside of 80; Maximin favors $A_3$'s minimum of 35. Minimax Regret and EMV/Bayes select $A_2$, using maximum regret 20 and EMV 50.5, respectively.

2. The minimum profits are:

$$A_1 : -10, \qquad A_2 : 20 + s, \qquad A_3 : 35.$$

   $A_1$ is ineligible because $-10 < 20$. Both $A_2$ and $A_3$ satisfy the minimum-profit requirement for all allowed values of $s$.

   For Maximin to prefer $A_2$ over $A_3$:

$$20 + s > 35 \quad \Longrightarrow \quad s > 15.$$

   Since $s$ must be an integer, the smallest subsidy is

$$\boxed{s = 16}.$$

   At $s = 16$, the minimum profits of the eligible alternatives are:

$$A_2 : 36, \qquad A_3 : 35.$$

   Maximin now recommends $A_2$ instead of $A_3$, since $36 > 35$. At $s = 15$, both alternatives tie at 35. The minimum subsidy for a strict change is therefore \$16,000.

   At $s = 16$, eligible payoffs are $(60, 55, 36)$ and $(40, 45, 35)$. Maxima are 60 and 45; minima are 36 and 35; maximum regrets are 0 and 20 (state bests $(60, 55, 36)$); EMVs are $50.5 + 0.20(16) = 53.7$ and 40.5. All four restricted criteria select $A_2$, with deciding values 60, 36, 0, and 53.7, respectively. Maximax changes from $A_1$ because of eligibility; Maximin changes from $A_3$ because the subsidy raises $A_2$'s worst profit above 35.

   **Updated recommendations at $s = 16$ (eligible alternatives only).**

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_2$ | Row maximum: 60 |
| Maximin | $A_2$ | Row minimum: 36 |
| Minimax Regret | $A_2$ | Maximum regret: 0 |
| EMV/Bayes | $A_2$ | EMV: 53.7 |



## 3.2 Business Reserve Investment

### Problem Description

A business invests part of its reserve for one year. The table gives *net gains*, in thousand US dollars, rather than final balances. All listed states are possible.

| Alternative | Growth ($S_1$) | Stable ($S_2$) | Recession ($S_3$) |
|---|---|---|---|
| $P(S_j)$ | 0.60 | 0.25 | 0.15 |
| Growth fund ($A_1$) | 90 | 25 | -30 |
| Balanced fund ($A_2$) | 60 | 40 | 5 |
| Fixed deposit ($A_3$) | 25 | 25 | 25 |

| Alternative | Row Maximum | Row Minimum | Maximum Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 90 | -30 | 55 | 55.75 |
| $A_2$ | 60 | 5 | 30 | 46.75 |
| $A_3$ | 25 | 25 | 65 | 25 |

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | _____ | _____ |
| Maximin | _____ | _____ |
| Minimax Regret | _____ | _____ |
| EMV/Bayes | _____ | _____ |

### Questions/Tasks



**C8** Which alternatives does each criterion recommend? Complete the answer table.

**C8** The business requires a gain of at least 5 in every state. A capital-protection contract adds $s$ to the balanced fund's recession payoff only; its cost is already included in the quoted payoffs. Here $s$ is an integer with $0 \leq s \leq 30$. Find the smallest $s$ that makes Maximin strictly prefer $A_2$ over $A_3$. Justify the threshold, identify eligible alternatives, and report all four restricted recommendations and deciding values at that $s$.

### Solution

1. The winning values in each summary column give:

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_1$ | Row maximum: 90 |
| Maximin | $A_3$ | Row minimum: 25 |
| Minimax Regret | $A_2$ | Maximum regret: 30 |
| EMV/Bayes | $A_1$ | EMV: 55.75 |

There are no ties. Maximax and EMV/Bayes both select $A_1$: one favors its maximum of 90, the other its EMV of 55.75. Their optimistic and risk-neutral priorities remain distinct; Maximin protects 25, while Minimax Regret limits missed opportunities to 30.

2. The minima are $-30$, $5+s$, and 25, since $5+s \leq 35 < 40$. Thus $A_1$ is excluded, while $A_2$ and $A_3$ remain eligible for every allowed $s$.

$$5+s > 25 \quad \Longrightarrow \quad s > 20.$$

The smallest integer is $s = 21$, a payment of \$21,000 in recession. At $s = 20$ Maximin ties at 25; at $s = 21$, $26 > 25$, so it changes from $A_3$ to $A_2$.

At $s = 21$, eligible payoffs are $(60, 40, 26)$ and $(25, 25, 25)$. Their maxima are 60 and 25; minima are 26 and 25; maximum regrets are 0 and 35 (state bests $(60, 40, 26)$); and EMVs are $46.75 + 0.15(21) = 49.9$ and 25. All four restricted criteria select $A_2$: Maximax uses 60, Maximin 26, Minimax Regret 0, and Bayes 49.9. Maximax and Bayes also change from $A_1$ because it fails the minimum-gain requirement. The protection raises the balanced fund's worst gain above the deposit's guaranteed gain.

**Updated recommendations at $s = 21$ (eligible alternatives only).**

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_2$ | Row maximum: 60 |
| Maximin | $A_2$ | Row minimum: 26 |
| Minimax Regret | $A_2$ | Maximum regret: 0 |
| EMV/Bayes | $A_2$ | EMV: 49.9 |



## 3.3 Coastal Ferry Capacity

### Problem Description

A ferry operator selects a seasonal capacity plan. Profits are in thousand US dollars. Seasonal booking forecasts give the probabilities below.

| Alternative | High ($S_1$) | Normal ($S_2$) | Low ($S_3$) |
|---|---|---|---|
| $P(S_j)$ | 0.40 | 0.40 | 0.20 |
| Add two ferries ($A_1$) | 100 | 35 | -25 |
| Add one ferry ($A_2$) | 75 | 55 | 10 |
| Lease peak capacity ($A_3$) | 55 | 48 | 30 |
| Keep current fleet ($A_4$) | 30 | 30 | 30 |

| Alternative | Row Maximum | Row Minimum | Maximum Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 100 | -25 | 55 | 49 |
| $A_2$ | 75 | 10 | 25 | 54 |
| $A_3$ | 55 | 30 | 45 | 47.2 |
| $A_4$ | 30 | 30 | 70 | 30 |

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | _____ | _____ |
| Maximin | _____ | _____ |
| Minimax Regret | _____ | _____ |
| EMV/Bayes | _____ | _____ |

**Questions/Tasks**



**C8** Which alternatives does each criterion recommend? Complete the answer table.

**C8** A booking guarantee replaces the two-ferry plan's low-demand payoff of $-25$ with $g$, an integer with $-25 \le g \le 30$. Other payoffs and probabilities stay unchanged; all four plans remain eligible. Find the smallest $g$ making $A_1$ the unique Minimax Regret choice. Justify the threshold, calculate the increase from $-25$, and report all four updated recommendations and deciding values at that $g$.

**Solution**

1. The winning values in each summary column give:

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_1$ | Row maximum: 100 |
| Maximin | $A_3$ and $A_4$ | Row minimum: 30 |
| Minimax Regret | $A_2$ | Maximum regret: 25 |
| EMV/Bayes | $A_2$ | EMV: 54 |

   Maximax favors the upside of 100, Maximin protects the minimum of 30, and Minimax Regret limits missed opportunities to 25. Report both $A_3$ and $A_4$ for Maximin: their minima match despite different upsides. Bayes selects $A_2$ because $54 > 49 > 47.2 > 30$.

2. The best payoffs in the three states remain $(100, 55, 30)$ for the allowed range of $g$. The regret rows and maximum regrets are

$$\begin{aligned} r(A_1) &= (0, 20, 30 - g), & R_1 &= \max\{20, 30 - g\}, \\ r(A_2) &= (25, 0, 20), & R_2 &= 25, \\ r(A_3) &= (45, 7, 0), & R_3 &= 45, \\ r(A_4) &= (70, 25, 0), & R_4 &= 70. \end{aligned}$$

   For a unique Minimax Regret choice,

$$\max\{20, 30 - g\} < 25 \quad \Longleftrightarrow \quad g > 5.$$

   At $g = 5$, $A_1$ and $A_2$ tie at 25. The smallest integer giving a strict change is $g = 6$, an increase of $6 - (-25) = 31$ thousand US dollars. The guarantee turns a loss of \$25,000 into a profit of \$6,000.
   At $g = 6$, all four plans remain eligible. Maxima are $(100, 75, 55, 30)$; minima are $(6, 10, 30, 30)$; maximum regrets are $(24, 25, 45, 70)$. Only $A_1$'s EMV changes: $49 + 0.20[6 - (-25)] = 55.2$; the others remain 54, 47.2, and 30. Maximax selects $A_1$ (maximum 100); Maximin selects $A_3$ and $A_4$ (minimum 30); Minimax Regret selects $A_1$

(maximum regret 24); Bayes selects $A_1$ (EMV 55.2). Regret and Bayes change from $A_2$ to $A_1$, by margins of 1 in maximum regret and 1.2 in expected profit.

**Updated recommendations at $g = 6$ (eligible alternatives only).**

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_1$ | Row maximum: 100 |
| Maximin | $A_3$ and $A_4$ | Row minimum: 30 |
| Minimax Regret | $A_1$ | Maximum regret: 24 |
| EMV/Bayes | $A_1$ | EMV: 55.2 |



## 3.4 Printing Workshop Equipment

### Problem Description

A printing workshop selects an equipment plan for the next year. Net profits are in thousand US dollars and already include all equipment costs.

| Alternative | High ($S_1$) | Normal ($S_2$) | Low ($S_3$) |
|---|---|---|---|
| $P(S_j)$ | 0.40 | 0.40 | 0.20 |
| Automated line ($A_1$) | 100 | 50 | 0 |
| Standard upgrade ($A_2$) | 75 | 55 | 25 |
| Maintain equipment ($A_3$) | 50 | 45 | 40 |

| Alternative | Row Maximum | Row Minimum | Maximum Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 100 | 0 | 40 | 60 |
| $A_2$ | 75 | 25 | 25 | 57 |
| $A_3$ | 50 | 40 | 50 | 46 |

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | _____ | _____ |
| Maximin | _____ | _____ |
| Minimax Regret | _____ | _____ |
| EMV/Bayes | _____ | _____ |

### Questions/Tasks



**C8** Which alternatives does each criterion recommend? Complete the answer table.

**C8** An additional annual service fee $f$ reduces the automated line's profit by $f$ in every state. Here $f$ is an integer with $0 \le f \le 5$; other payoffs and probabilities stay unchanged. The workshop requires EMV of at least 55. Find the smallest $f$ making Bayes strictly prefer $A_2$ over $A_1$. Justify the threshold, identify eligible plans throughout the range, and report all four restricted recommendations and deciding values at that fee. Deduct the fee once.

**Solution**

1. The winning values in each summary column give:

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_1$ | Row maximum: 100 |
| Maximin | $A_3$ | Row minimum: 40 |
| Minimax Regret | $A_2$ | Maximum regret: 25 |
| EMV/Bayes | $A_1$ | EMV: 60 |

   There are no ties. Maximax and EMV/Bayes select $A_1$ for its maximum of 100 and EMV of 60. Maximin protects a minimum of 40 with $A_3$; Minimax Regret selects $A_2$ to limit missed opportunities to 25.

2. Updated payoffs for $A_1$ are $(100-f, 50-f, -f)$. Since probabilities sum to one, its EMV is $60-f$; the other EMVs remain 57 and 46. Eligibility requires $60-f \ge 55$, or $f \le 5$. Thus $A_1$ and $A_2$ remain eligible throughout the allowed range; $A_3$ fails because $46 < 55$.

$$57 > 60 - f \quad \Longleftrightarrow \quad f > 3.$$

   The smallest integer is $f = 4$, an annual fee of \$4,000. At $f = 3$ Bayes ties at 57; at $f = 4$, $A_2$ wins by $57 - 56 = 1$ thousand US dollars in expected profit.
   Eligible payoffs at $f = 4$ are $(96, 46, -4)$ and $(75, 55, 25)$. Maxima are 96 and 75; minima are $-4$ and 25; EMVs are 56 and 57. State bests $(96, 55, 25)$ give regret rows $(0, 9, 29)$ and $(21, 0, 0)$, with maximum regrets 29 and 21. Restricted Maximax selects $A_1$ (maximum 96); Maximin selects $A_2$ (minimum 25); Minimax Regret selects $A_2$ (maximum regret 21); Bayes selects $A_2$ (EMV 57). Bayes changes from $A_1$ because the fee removes its expected-profit advantage. Maximin changes from $A_3$ because that plan fails the EMV requirement. Two plans remain eligible even at $f = 5$.
   **Updated recommendations at $f = 4$ (eligible alternatives only).**

| Criterion | Recommended Alternative(s) | Deciding Value |
|---|---|---|
| Maximax | $A_1$ | Row maximum: 96 |
| Maximin | $A_2$ | Row minimum: 25 |
| Minimax Regret | $A_2$ | Maximum regret: 21 |
| EMV/Bayes | $A_2$ | EMV: 57 |



# 4 Mixed Practice

**Problems:**


4.1. Choosing a criterion

4.2. Evaluating decision claims

4.3. Decision committee report


## 4.1 Choosing a Criterion

Which criterion fits each *stated priority*? Give its required information and one limitation, and briefly contrast another feasible criterion.

1. A start-up knows its payoffs but lacks reliable probabilities. Which criterion suits a founder seeking maximum upside and accepting the downside?
2. A family business has reliable probabilities. Which criterion suits its priority of the highest worst-case profit?
3. A delivery manager lacks reliable probabilities. Which criterion minimizes the greatest profit forgone relative to the best plan in each state?
4. A retailer trusts its probabilities and makes repeated seasonal decisions. Which criterion suits its priority of average monetary profit?
5. A museum knows its payoffs but lacks attendance probabilities. What information is missing for an EMV choice, and why would equal probabilities be an assumption?

**Solution**

1. Maximax fits the optimistic priority and needs only payoffs, but ignores downside outcomes. Maximin is also feasible and instead protects the worst payoff.
2. Maximin fits the conservative priority and needs only payoffs, but ignores upside and likelihoods. EMV/Bayes is feasible but prioritizes expected profit.
3. Minimax Regret needs the full payoff table, without probabilities, but may sacrifice worst-case protection. Maximin is feasible and instead protects the minimum profit.
4. EMV/Bayes fits the risk-neutral priority and needs payoffs and reliable probabilities, but cannot guarantee a season's profit. Maximin is feasible and instead protects the worst outcome.
5. A nonnegative probability for each attendance state, summing to one, is missing. Equal probabilities would assume equally likely states without evidence. The three payoff criteria remain feasible but answer different priorities.



## 4.2 Evaluating Decision Claims

Is each claim valid? Correct it if needed, using a numerical example where possible.

1. "Maximin and Minimax Regret always agree because both are cautious."
2. "An EMV of 55 guarantees a seasonal profit of 55."
3. "Regret of 25 means an actual monetary loss of 25."
4. "Agreement with Maximax makes Bayes optimistic."

5. "Valid probabilities make EMV best for every decision-maker."

**Solution**

1. Invalid. In delivery expansion, Maximin selects $A_3$ (minimum 35); Minimax Regret selects $A_2$ (maximum regret 25, minimum 10). Worst-case protection and limited missed opportunities are different priorities.
2. Invalid. Delivery plan $A_4$ has EMV 55 but can earn only 60, 65, or 20. Expected profit is not guaranteed profit.
3. Invalid. With low festival attendance, $A_1$ earns 10 but has regret $35 - 10 = 25$. Regret measures forgone profit, not an actual loss.
4. Invalid. Both choose investment $A_1$, but Maximax uses its maximum of 90 and Bayes its EMV of 55.75. Bayes remains risk-neutral.
5. Invalid. The farm's EMV choice $A_2$ has minimum 20 and fails the required 30. Only $A_3$, with minimum 35 and EMV 40.5, meets the obligation despite valid probabilities.



## 4.3 Decision Committee Report

Return to the delivery expansion problem in Section 2.1. Four committee members prioritize, respectively, the largest possible profit, the best worst-case profit, the smallest maximum missed opportunity, and the highest expected profit.

1. Which criterion and plan fit each member's priority? Give the deciding value and information used or ignored.
2. Which plan would you recommend? Compare two plans numerically, state your priority and one limitation, and explain a reasonable disagreement.
3. If the forecast is withdrawn, which summaries remain usable and which recommendation loses support? Does one remaining criterion become universally best?

**Solution**

1. In the stated order: Maximax chooses $A_1$ (row maximum 100), ignoring downside and probabilities; Maximin chooses $A_3$ (row minimum 35), ignoring upside and probabilities. Minimax Regret chooses $A_2$ (maximum regret 25), comparing all payoffs but ignoring likelihoods; EMV/Bayes chooses $A_4$ (EMV 55), using payoffs and probabilities without protecting the worst outcome.
2. One valid report is: "We prioritize expected profit and recommend $A_4$: its EMV is 55 versus 47 for $A_2$. This depends on reliable probabilities and does not guarantee 55. A member prioritizing missed opportunities may prefer $A_2$, with maximum regret 25 versus 40 for $A_4$." Other choices are valid with a stated priority, numerical support, and a limitation.
3. Row maxima, row minima, and maximum regrets remain usable, retaining choices $A_1$, $A_3$, and $A_2$, respectively. The EMVs and Bayes choice $A_4$ lose their forecast basis. No remaining criterion becomes universally best: each serves a different priority.

**Self-check.** State the information, priority, recommendations (including ties), deciding values, and limitations.

